

UNIT 4

Special Continuous Distributions

[Normal, Gamma and Exponential]

Normal Distribution

[Also known as the **Gaussian distribution**, in honour of Karl Friedrich Gauss (1777–1855)]

A continuous random variable X with a bell-shaped Figure distribution is called a **normal random variable**. The mathematical equation for the probability distribution of the normal variable depends on the two parameters μ and σ , its **mean** and **standard deviation**, respectively. Hence, we denote the values of the density of X by $n(x; \mu, \sigma)$ or $N(\mu, \sigma)$

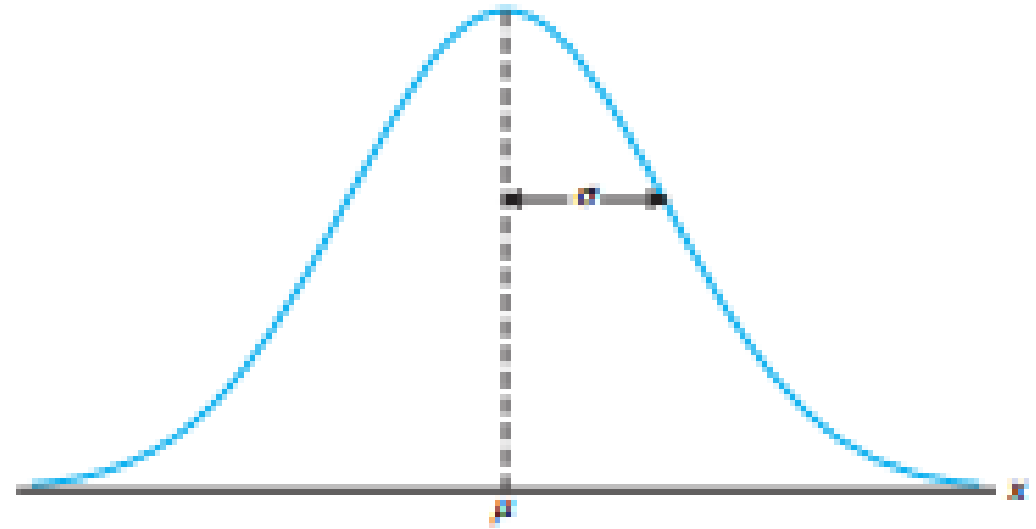


Figure 1.1 : The normal curve.

The probability density function of normal distribution

The density of the normal random variable X , with mean μ and variance σ^2 , is

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty,$$

where $\pi = 3.14159\dots$ and $e = 2.71828\dots$

Properties of Normal distribution

1. The mode, the point on the horizontal axis where the curve is a maximum, occurs at $x = \mu$.
2. The curve is symmetric about a vertical axis through the mean μ .
3. The curve has its points of inflexion at $x = \mu \pm \sigma$; it is concave downward if $\mu - \sigma < X < \mu + \sigma$ and is concave upward otherwise.
4. The normal curve approaches the horizontal axis asymptotically as we proceed in either direction away from the mean.
5. **The total area under the curve and above the horizontal axis is equal to 1.**
6. The mean and variance of $n(x; \mu, \sigma)$ are μ and σ^2 , respectively. Hence, the standard deviation is σ .

A point of inflexion is a point on a smooth curve where the curvature changes sign.

Normal distribution is also known as Gaussian distribution or Bell-shaped Curve distribution

Properties of Normal distribution:

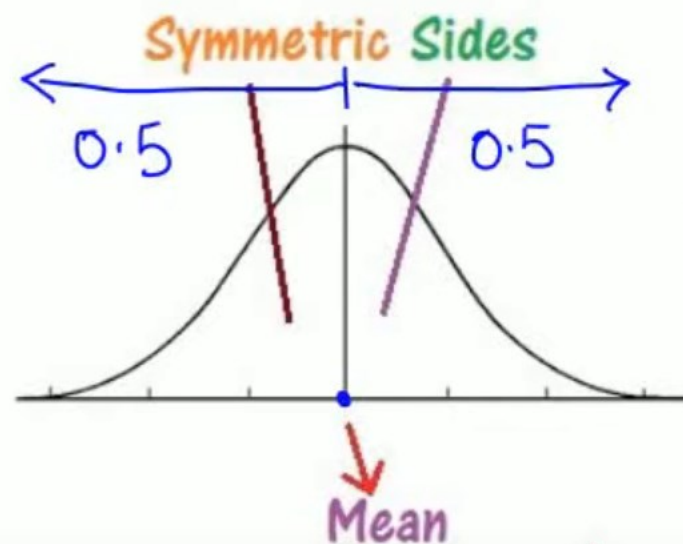
1) The Mean, median and Mode are all EQUAL

2) The curve is symmetric about the mean.

3) Total area under the curve is 1.

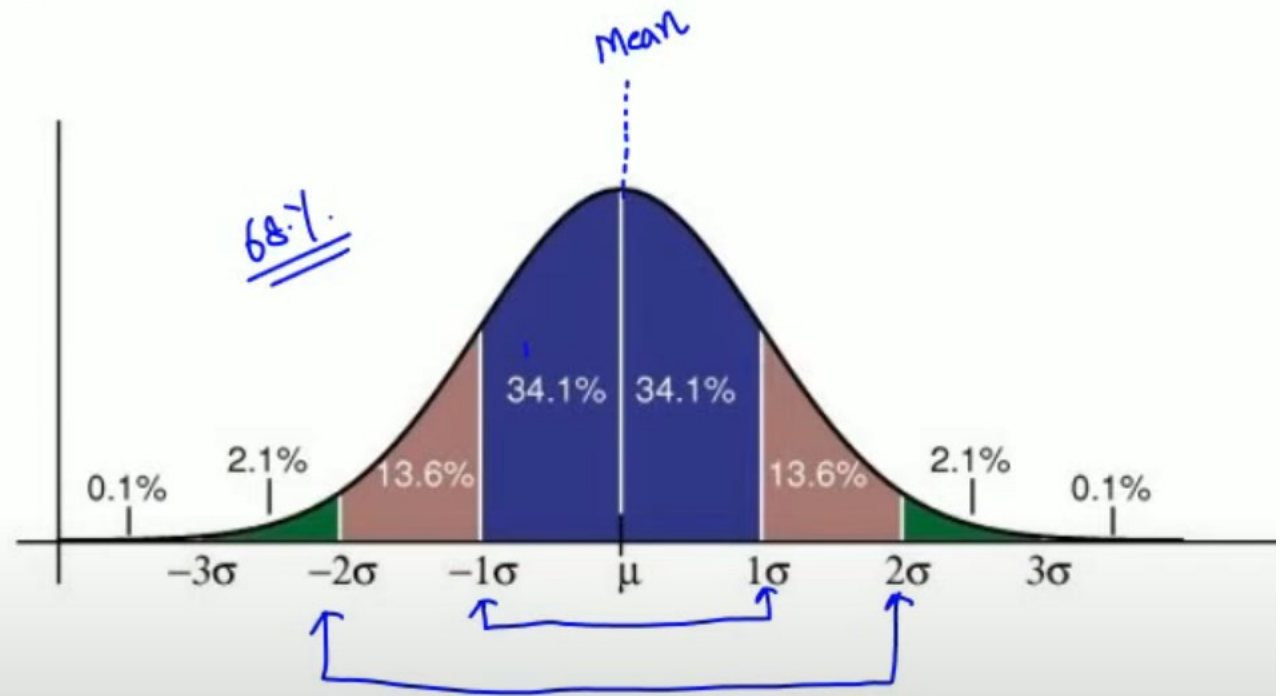
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

4) Area to the left and area to right about the mean are same i.e., 0.5



The **empirical rule** tells you what **percentage of your** data falls within a certain number of standard deviations (σ) from the mean (μ):

- ✓ **68%** of the data falls within **one** σ of the μ .
- **95%** of the data falls within **two** σ of the μ .
- **99.7%** of the data falls within **three** σ of the μ .



Areas under the Normal Curve

The curve of any continuous probability distribution or density function is constructed so that the area under the curve bounded by the two ordinates $x = x_1$ and $x = x_2$ equals the probability that the random variable X assumes a value between $x = x_1$ and $x = x_2$. Thus, for the normal curve in Figure 6.6,

$$P(x_1 < X < x_2) = \int_{x_1}^{x_2} n(x; \mu, \sigma) dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{x_1}^{x_2} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$

is represented by the area of the shaded region.

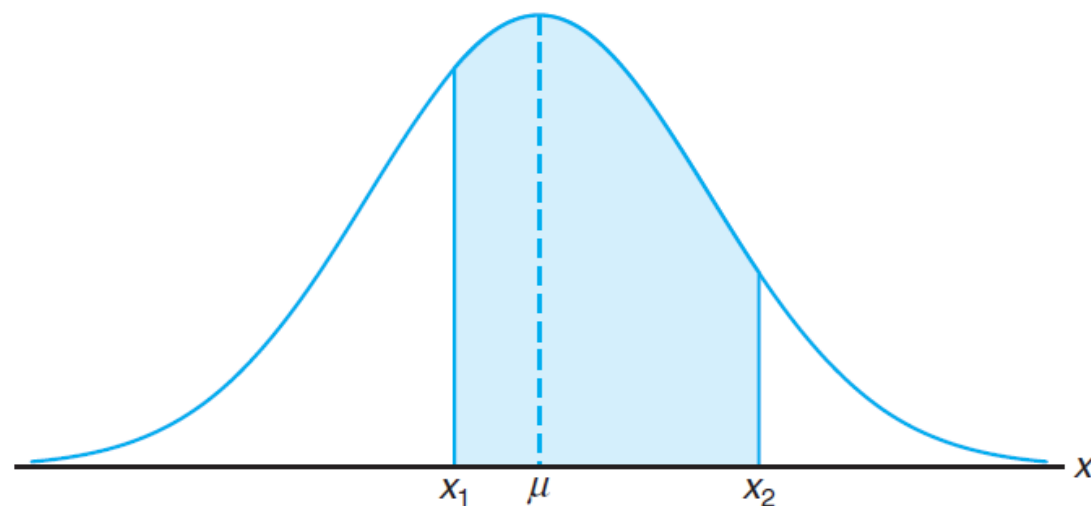


Figure 6.6: $P(x_1 < X < x_2) = \text{area of the shaded region.}$

Standard Normal variate

A normal random variable distribution with a mean of 0 and variance of 1 is called a *standard normal distribution*.

This can be done by means of the transformation $Z = \frac{X - \mu}{\sigma}$.

The probability density function of Z is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} ; -\infty < Z < \infty$$

Mean of Z is

$$\begin{aligned} E(Z) &= E\left(\frac{X - \mu}{\sigma}\right) \\ &= \frac{1}{\sigma} E(X - \mu) \\ &= \frac{1}{\sigma} [E(X) - E(\mu)] \\ &= \frac{1}{\sigma} [\mu - \mu] \\ &= 0 \end{aligned}$$

$V(X+a) = V(X)$

$X \sim N(\mu, \sigma^2)$
 $Z \sim N(0, 1)$

Variance of Z is

$$\begin{aligned} &= \text{Var}\left(\frac{X - \mu}{\sigma}\right) \\ &= \frac{1}{\sigma^2} \text{Var}(X - \mu) \\ &= \frac{1}{\sigma^2} [\text{Var}(X) + \text{Var}(\mu)] \\ &= \frac{1}{\sigma^2} [\sigma^2 + 0] \\ &= 1 \end{aligned}$$

Transformation (Normal distribution to Standard normal distribution)

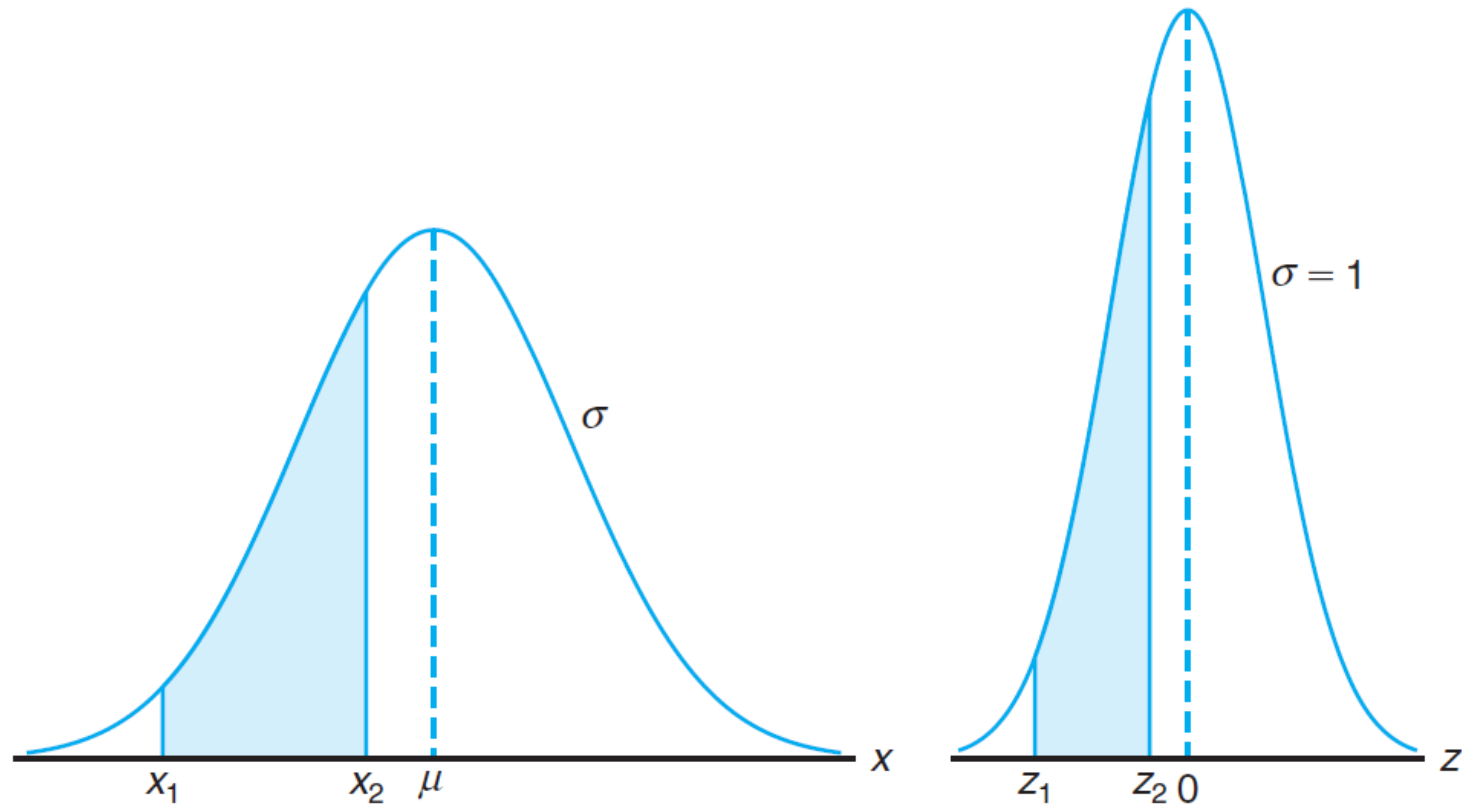


Figure 6.8: The original and transformed normal distributions.

Moment generating function (m.g.f) of Normal Distribution

- The moment-generating function of the normal distribution;

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

MGF for standard random variable $Z \sim N(0,1)$

$$M_Z(t) = e^{\frac{t^2}{2}}$$

Find mean and Variance using MGF

Example: An electrical firm manufactures light bulbs that have life, before burn-out, that is normally distributed with mean is 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns (i) more than 834 hours (ii) between 778 and 834 hours.

Given that $P(Z < -0.85) = 0.1977$; $P(0 < Z < 0.85) = 0.3023$; $P(-0.55 < Z < 0) = 0.2088$; $P(Z < 0.85) = 0.8023$.

Example: The saving **bank account of** a customer showed an **average balance** of \$150 and a **standard deviation** of \$50. Assuming that the **account balances are** normally distributed, find **what percentage of** account is (i) over \$200 (ii) between \$120 and \$170 (iii) less than \$75. Given that $P(0 < Z < 1) = 0.3413$; $P(Z < 0.4) = 0.6554$; $P(Z < -0.6) = 0.2743$; $P(0 < Z < 1.5) = 0.4332$.

3. Given a standard normal distribution, find the area under the curve that lies
- (a) to the right of $z = 1.84$ and
 - (b) between $z = -1.97$ and $z = 0.86$.

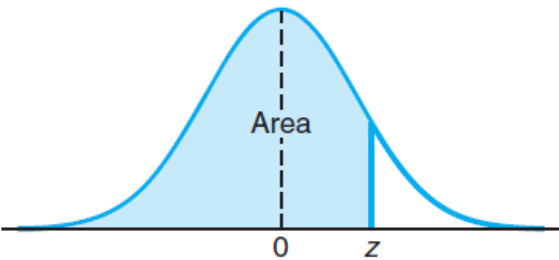


Table A.3 Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143

−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

4. Given a standard normal distribution, find the value of k such that

(a) $P(Z > k) = 0.3015$ and

(b) $P(k < Z < -0.18) = 0.4197$.

- 5. Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.**
- 6. A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.**

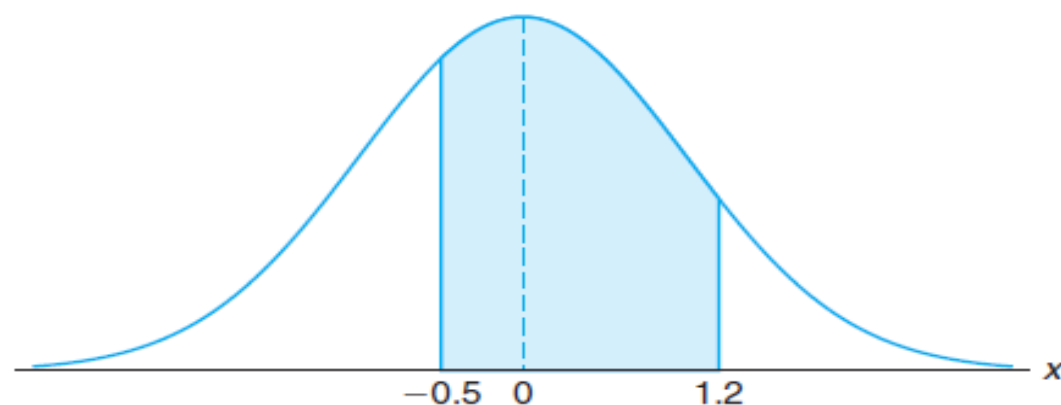


Figure 6.11: Area for Example 6.4.

Solution: The z values corresponding to $x_1 = 45$ and $x_2 = 62$ are

$$z_1 = \frac{45 - 50}{10} = -0.5 \text{ and } z_2 = \frac{62 - 50}{10} = 1.2.$$

Therefore,

$$P(45 < X < 62) = P(-0.5 < Z < 1.2).$$

$P(-0.5 < Z < 1.2)$ is shown by the area of the shaded region in Figure 6.11. This area may be found by subtracting the area to the left of the ordinate $z = -0.5$ from the entire area to the left of $z = 1.2$. Using Table A.3, we have

$$\begin{aligned} P(45 < X < 62) &= P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5) \\ &= 0.8849 - 0.3085 = 0.5764. \end{aligned}$$



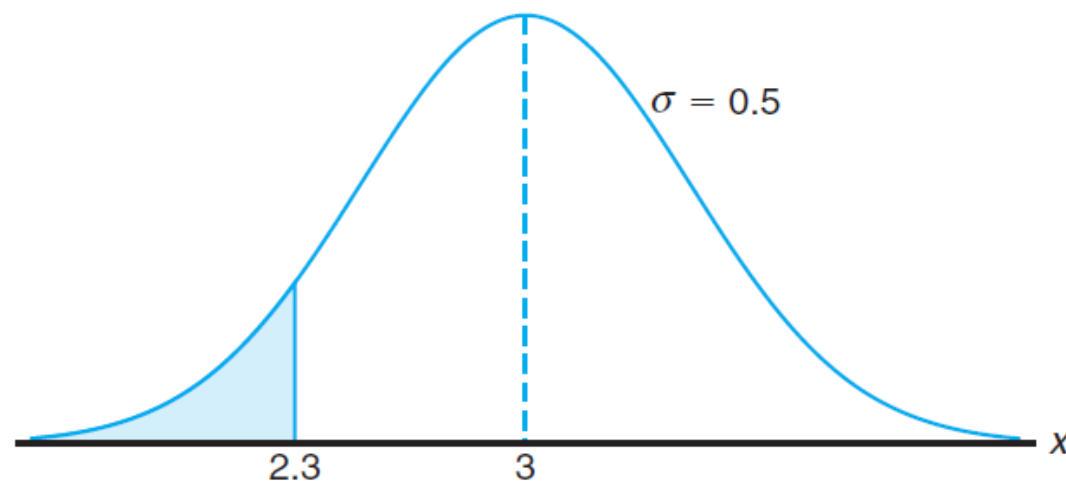


Figure 6.14: Area for Example 6.7.

First construct a diagram such as Figure 6.14, showing the given distribution of battery lives and the desired area. To find $P(X < 2.3)$, we need to evaluate the area under the normal curve to the left of 2.3. This is accomplished by finding the area to the left of the corresponding z value. Hence, we find that

$$z = \frac{2.3 - 3}{0.5} = -1.4,$$

and then, using Table A.3, we have

$$P(X < 2.3) = P(Z < -1.4) = 0.0808.$$



Using the Normal Curve in Reverse

Sometimes, we are required to find the value of z corresponding to a specified probability that falls between values listed in Table. For convenience, we shall always choose the z value corresponding to the tabular probability that comes closest to the specified probability.

We will go first from a value of x to a z value and then compute the desired area.

In Example 1, we reverse the process and begin with a known area or probability, find the z value, and then determine x by rearranging the formula

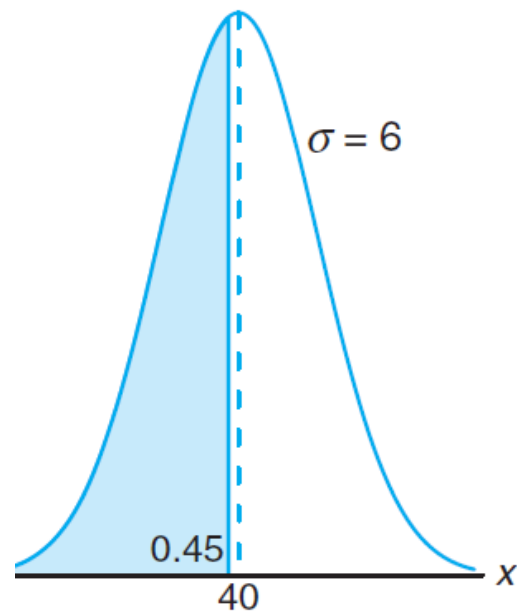
$$z = (x - \mu) / \sigma \text{ to give } x = \sigma z + \mu.$$

Example: Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value of x that has

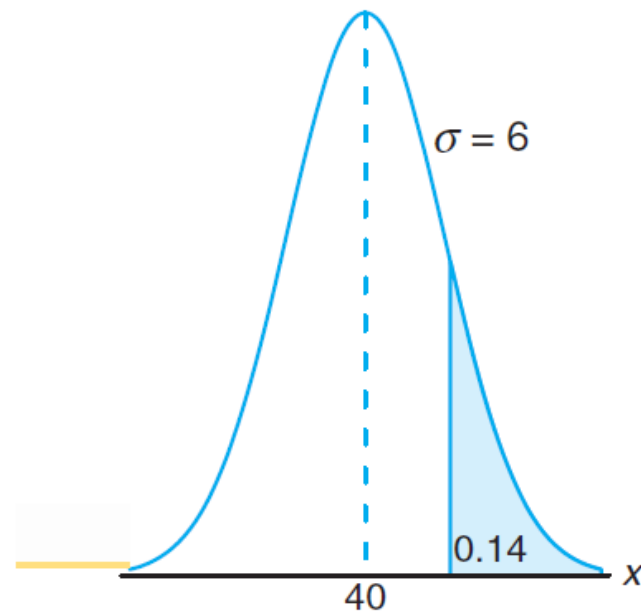
- (a) 45% of the area to the left and
- (b) 14% of the area to the right.

Solution:

- (a) An area of 0.45 to the left of the desired x value is shaded in Figure. We require a z value that leaves an area of 0.45 to the left. From Table we find $P(Z < -0.13) = 0.45$, so the desired z value is -0.13 . Hence, $x = (6)(-0.13) + 40 = 39.22$.
- (b) In Figure, we shade an area equal to 0.14 to the right of the desired x value. This time we require a z value that leaves 0.14 of the area to the right and hence an area of 0.86 to the left. Again, from Table A.3, we find $P(Z < 1.08) = 0.86$, so the desired z value is 1.08 and $x = (6)(1.08) + 40 = 46.48$



(a)



(b)

PRACTICE QUESTIONS

6.5 Given a standard normal distribution, find the area under the curve that lies

- (a) to the left of $z = -1.39$;
- (b) to the right of $z = 1.96$;
- (c) between $z = -2.16$ and $z = -0.65$;
- (d) to the left of $z = 1.43$;
- (e) to the right of $z = -0.89$;
- (f) between $z = -0.48$ and $z = 1.74$.

Given a standard normal distribution, find the value of k such that

(a) $P(Z > k) = 0.2946$;

(b) $P(Z < k) = 0.0427$;

(c) $P(-0.93 < Z < k) = 0.7235$.

Given that X has a normal distribution with $\mu = 300$ and $\sigma = 50$, find the probability that X assumes a value greater than 362.

Given a normal distribution with $\mu = 30$ and $\sigma = 6$, find

- (a) the normal curve area to the right of $x = 17$;
- (b) the normal curve area to the left of $x = 22$;
- (c) the normal curve area between $x = 32$ and $x = 41$;
- (d) the value of x that has 80% of the normal curve area to the left;
- (e) the two values of x that contain the middle 75% of the normal curve area

Let X is normally distributed and mean of X is 12 and standard deviation is 4. Find out the probability of the following
a) $X \geq 20$, b) $X \leq 20$ and c) $0 \leq X \leq 12$.

Sol:

Find the value of z if the area under a standard normal curve

(a) to the right of z is 0.3622;

(b) to the left of z is 0.1131;

(c) between 0 and z , with $z > 0$, is 0.4838;

(d) between $-z$ and z , with $z > 0$, is 0.9500

Given the normally distributed variable X with a mean 18 and standard deviation of 2.5, find

- (a) $P(X < 15)$;
- (b) the value of k such that $P(X < k) = 0.2236$;
- (c) the value of k such that $P(X > k) = 0.1814$;
- (d) $P(17 < X < 21)$

Let X is normally distributed and mean of X is 30 and standard deviation is 5. Find out the probability of the following

- a) $26 \leq X \leq 40$ b) $X \geq 45$ and c) $|X - 30| > 5$.

The heights of 1000 students are normally distributed with a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Assuming that the heights are recorded to the nearest half-centimeter, how many of these students would you expect to have heights

- (a) less than 160.0 centimeters?
- (b) between 171.5 and 182.0 centimeters inclusive?
- (c) equal to 175.0 centimeters?
- (d) greater than or equal to 188.0 centimeters?

If a set of observations is normally distributed, what percent of these differ from the mean by

- (a) more than 1.3σ ?
- (b) less than 0.52σ ?

Solved questions

1. An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

Solution: The z values corresponding to $x_1 = 778$ and $x_2 = 834$ are

$$z_1 = \frac{778 - 800}{40} = -0.55 \text{ and } z_2 = \frac{834 - 800}{40} = 0.85.$$

Hence,

$$\begin{aligned} P(778 < X < 834) &= P(-0.55 < Z < 0.85) = P(Z < 0.85) - P(Z < -0.55) \\ &= 0.8023 - 0.2912 = 0.5111. \end{aligned}$$

2. A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.

Solution: To find $P(X < 2.3)$, we need to evaluate the area under the normal curve to the left of 2.3. This is accomplished by finding the area to the left of the corresponding z value. Hence, we find that

$$z = \frac{2.3 - 3}{0.5} = -1.4,$$

and then, using Table A.3, we have

$$\underline{P(X < 2.3)} = P(Z < -1.4) = 0.0808.$$

3. A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live

- (a) more than 32 months;
- (b) less than 28 months;
- (c) between 37 and 49 months

To solve these probability problems involving normally distributed lifetimes of mice, we can use the standard normal distribution (Z-distribution). The mean (μ) is 40 months, and the standard deviation (σ) is 6.3 months. We can convert the given lifetimes to Z-scores using the formula:

$$Z = \frac{X - \mu}{\sigma}$$

where X is the value we are interested in.

(a) Probability that a given mouse will live more than 32 months

1. Calculate the Z-score for 32 months:

$$Z = \frac{32 - 40}{6.3} \approx \frac{-8}{6.3} \approx -1.27$$

2. Look up the Z-score in the standard normal distribution table or use a calculator to find $P(Z < -1.27)$:

$$P(Z < -1.27) \approx 0.1020$$

3. Therefore, the probability that a mouse lives **more than 32 months** is:

$$P(X > 32) = 1 - P(Z < -1.27) \approx 1 - 0.1020 \approx 0.8980$$

(b) Probability that a given mouse will live less than 28 months

1. Calculate the Z-score for 28 months:

$$Z = \frac{28 - 40}{6.3} \approx \frac{-12}{6.3} \approx -1.90$$

2. Look up the Z-score in the standard normal distribution table or use a calculator:

$$P(Z < -1.90) \approx 0.0287$$

3. Thus, the probability that a mouse lives **less than 28 months** is:

$$P(X < 28) \approx 0.0287$$

(c) Probability that a given mouse will live between 37 and 49 months

1. Calculate the Z-scores for 37 and 49 months:

- For 37 months:

$$Z = \frac{37 - 40}{6.3} \approx \frac{-3}{6.3} \approx -0.48$$

Look up the Z-score:

$$P(Z < -0.48) \approx 0.3186$$

- For 49 months:

$$Z = \frac{49 - 40}{6.3} \approx \frac{9}{6.3} \approx 1.43$$

Look up the Z-score:

$$P(Z < 1.43) \approx 0.9236$$

2. Thus, the probability that a mouse lives **between 37 and 49 months** is:

$$P(37 < X < 49) = P(Z < 1.43) - P(Z < -0.48) \approx 0.9236 - 0.3186 \approx 0.6050$$

A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,

- (a) what fraction of the cups will contain more than 224 milliliters?
- (b) what is the probability that a cup contains between 191 and 209 milliliters?
- (c) how many cups will probably overflow if 230- milliliter cups are used for the next 1000 drinks?
- (d) below what value do we get the smallest 25% of the drinks?

Normal Approximation to the Binomial

If X is a binomial random variable with mean $\mu = np$ and variance $\sigma^2 = npq$, then the limiting form of the distribution of

$$Z = \frac{X - np}{\sqrt{npq}},$$

as $n \rightarrow \infty$, is the standard normal distribution $n(z; 0, 1)$.

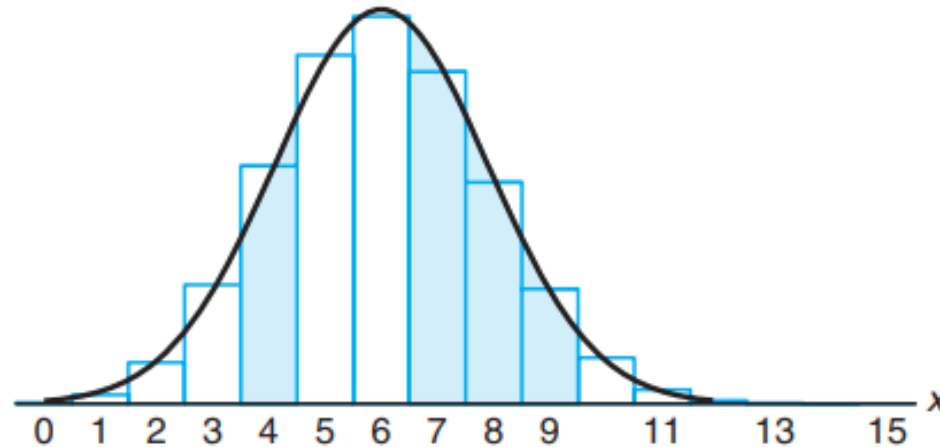


Figure 6.23: Normal approximation of $b(x; 15, 0.4)$ and $\sum_{x=7}^9 b(x; 15, 0.4)$.

if we seek the area under the normal curve to the left of, say, x , it is more accurate to use $x + 0.5$. This is a correction to accommodate the fact that a discrete distribution is being approximated by a continuous distribution. The correction **+0.5** is called a **continuity correction**.

Example 1

6 The probability that a patient recovers from a delicate heart operation is 0.9. Of the next 100 patients having this operation, what is the probability that

(a) between 84 and 95 inclusive survive?

(b) fewer than 86 survive?

Example 2

A multiple-choice quiz has 200 questions, each with 4 possible answers of which only 1 is correct. What is the probability that sheer guesswork yields from 25 to 30 correct answers for the 80 of the 200 problems about which the student has no knowledge?

PRACTICE QUESTIONS

1 If 20% of the residents in a U.S. city prefer a white telephone over any other color available, what is the probability that among the next 1000 telephones installed in that city

- (a) between 170 and 185 inclusive will be white?
- (b) at least 210 but not more than 225 will be white?

2. The probability that a patient recovers from a rare blood disease is 0.4. If 100 people are known to have contracted this disease, what is the probability that fewer than 30 survive?

3. If 20% of the residents in a U.S. city prefer a white telephone over any other color available, what is the probability that among the next 1000 telephones installed in that city

- (a) between 170 and 185 inclusive will be white?
- (b) at least 210 but not more than 225 will be white?

MCQS on Normal Distributions

1. **Standard normal distribution has which of the following properties**

- a) Mean=Variance=1
- b) Standard deviation=Variance=0
- c) Mean=0, Variance=1
- d) Mean=standard deviation

2. **If $X \sim N(1, 5)$ the probability density function for the normal variate X is**

- a) $\frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-1}{5}\right)^2}$ b) $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-1}{5}\right)^2}$ c) $\frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ d) $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2}$

3. **If m.g.f of normal variate is $M_X(t) = e^{\left(\frac{24t+16t^2}{2}\right)}$ then mean(μ) and variance(σ^2) of X are**

- a) $\mu=24$ $\sigma^2=4$ b) $\mu=12$ $\sigma^2=4$
c) $\mu=12$ $\sigma^2=16$ d) $\mu=24$ $\sigma^2=16$

4. **If $X \sim N(65, 25^2)$ with mean and variance as parameters, then value of standardized normal variable for $X = 35$ is**

- (A) -1.8 (B) -1.6 (C) -1.4 (D) -1.2

MCQS on Normal Distributions (Gate exam)

1. A normal random variable X has the following probability density function

$$f(x) = \frac{1}{\sqrt{8\pi}} e^{-\left[\frac{(x-1)^2}{8}\right]}, \quad -\infty < x < \infty. \text{ Then } \int_1^{\infty} f(x) dx =$$

- a) 0 b) $\frac{1}{2}$ c) $1 - \frac{1}{e}$ d) 1

2. Let X be a normal random variable with mean zero and variance 9. If $a = P(X \geq 3)$, then $P(|X| \leq 3)$ equals to

- a) a b) $2a$ c) $1-2a$ d) $1-a$

3. If X is a normal variate with mean 30 and standard deviation 5, what is the probability $P(26 \leq X \leq 34)$?

Given $A(Z > 0.8) = 0.2118$

- a) 0.2118 b) 0.4236 c) 0.7882 d) 0.5764

4. Let X be a random variable following normal distribution with mean +1 and variance 4. Let Y be another normal variable with mean -1 and variance unknown.

If $P(X \leq -1) = P(Y \geq 2)$, the standard deviation of Y is

- a) 3 b) 2 c) $\sqrt{2}$ d) 1